Shivam Saxena

Lead Data Engineer | Data Architecture | Snowflake | Python | SQL | AI/ML Solutions
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Professional Summary

Lead Data Engineer with 10 years of experience designing and delivering data-driven solutions across Snowflake, Oracle, Python, and enterprise analytics platforms. Experienced in leading teams of up to 6 engineers, driving requirements and data modeling, owning solution architecture and LLDs, and delivering scalable data pipelines and reporting platforms. Hands-on experience in applied AI/NLP solutions, data modernization, automation, and cross-functional delivery across banking, payments, marketing, and compliance use cases.

Certifications — SnowPro Core Certification – Snowflake | 2026 - 2028

Education

Bachelors of Engineering — Computer Science and Engineering
Galgotias College of Engineering and Technology, Greater Noida, UP | 2016

Professional Experience

Lead Data Engineer

Netspend India LLP | Noida | Apr 2025 – Present

Led a team of 6 Analytics Engineers, providing technical direction across data architecture, pipeline development, analytics solutions, and delivery of business-critical initiatives.

- **PFG Accounts Migration & Pipeline Architecture**
  - Led the data engineering workstream for a strategic PFG account migration covering 700K customer accounts and 1.5M cards, delivered within a 2-month timeline.
  - Collaborated with external partners to define data requirements, data models, migration approach, and data-sharing architecture.
  - Designed and implemented scalable Snowflake data pipelines and transformation logic to support the migration and downstream business requirements.
  - Implemented **Snowflake Secure Data Sharing** with **Secure Views** to provide external partners real-time access to required data while preventing exposure of internal business tables, transformation logic, and underlying implementation details.
- **Playstation Subscription & Rewards Program Setup**
  - Designed data validation logic to continuously assess customer subscription health and account standing for PlayStation users.
  - Automated identification of cashback-eligible accounts and downstream data handoffs to Finance for wallet-credit processing.
- **AI Marketing Collateral Tool**
  - Led development of an AI-powered compliance tool using Claude on AWS Bedrock and Streamlit to scan marketing collateral for regulatory trigger phrases and map them to relevant disclosures.
  - Reduced legal review turnaround time by 40% and enabled the Marketing team to perform self-service compliance checks.
- **Reconciliation and Audit Reports**
  - Designed and implemented SQL + Python solutions for **bank reconciliation, partner commission calculations, third-party audit reporting, US INT-1099 reporting, and Visa/Mastercard/Fiserv/Pulse network reconciliation**.
  - Built Python-based parallel processing capabilities to accelerate report generation and handle large-scale reporting workloads.
  - Optimized the commission reporting workflow to generate reports for **700+ partners in under 10 minutes**, significantly improving reporting turnaround time.
- **Custom Customer Statement Generation Platform**
  - Designed and developed a **Python FastAPI service**, containerized with Docker, to generate customized customer statements on demand.
  - Integrated the service with the company's primary application, accepting customer number, email address, and statement date range as request parameters.
  - Implemented automated statement generation and email delivery, completing customer requests within **5–10 minutes**.

- Architected the solution as a **back-office workload**, offloading statement-generation processing from the core transactional environment and reducing computational load on the primary application.

- **ACH Company Name Matching Automation (NLP + Fuzzy Logic)**
  - Designed and implemented an entity-matching solution to map ACH company names with cardholder employee data using hybrid similarity scoring combining fuzzy matching and transformer-based NLP embeddings.
  - Reduced Banking Operations' manual review workload from **100+ cases per day to approximately 10–15 cases per day**, with remaining reviews primarily focused on **first-time company names not previously encountered by the matching system**.
  - Automated the majority of recurring reconciliation matches, reducing operational delays in banking operations and money-movement processes.
- **Dockerized Python Reporting Framework Modernization**
  - I solved a production-scale architecture problem by Re-architected Python reporting framework from single-instance deployment into containerized (Docker-based) architecture enabling isolated dependency management per report.

Associate Lead Data Engineer

Global Payments Solutions | Noida | Oct 2022 – Mar 2025

Led a team of 6 Data Engineers delivering enterprise data engineering, analytics, and reporting solutions.

- **Enterprise AML (Actimize) Analytics Platform**
  - Led requirements gathering and data modeling for 66 analytics feed tables supporting enterprise-wide AML transaction monitoring.
  - Designed detailed LLDs for Snowflake ELT pipelines, leveraging dynamic tables, tasks, and Python stored procedures.
- **Customer Marketing Analytics - Braze**
  - Architected a Snowflake-based customer analytics feed supporting **1.5M accounts with 33 customer attributes per account**, supplying customer data to Braze for marketing decisioning.
  - Designed an incremental JSON-based delivery approach that sends **only changed attributes for each account**, minimizing redundant data transfer and improving downstream feed efficiency.
- **Secure Credit Card Analytics**
  - Designed and implemented a verification framework for a newly introduced company-wide credit logic, validating **available balance, current balance, and credit limit** against transaction activity and ISO8583 message data.
  - Developed transaction-level analytics and validation logic to identify discrepancies and provide confidence in the accuracy of credit-balance calculations.
- **Talend ETL Modernization & Python Migration**
  - Led migration of large-scale Talend ETL workflows to scalable Python-based data pipelines, **eliminating dependency on licensed ETL tooling and saving approximately $100K in licensing costs**.
  - Architected a modular transformation framework supporting multi-source ingestion, complex business rules, and high-volume data processing.
- **Python Analytics Reporting Framework**
  - Architected a config-driven Python analytics reporting framework supporting Snowflake, Oracle, and Postgres.

Senior Analyst Programmer (Analytics Engineering)

Fidelity International | Gurgaon | Apr 2022 – Oct 2022

- Built Python-based data ingestion and transformation pipelines using Pandas for analytical use cases.
- Partnered with HR stakeholders to design and automate employee vitality metrics and analytics datasets.

Software Engineer (Data & Analytics Systems)

TSYS Card Tech Ltd | Noida | May 2019 – Apr 2022

- Designed and optimized Oracle SQL procedures for billing, commissions, and financial settlement analytics.
- Developed complex settlement logic using PL/SQL collections and advanced SQL patterns.

Programmer Analyst

Cognizant Technology Solutions | Chennai | Jun 2016 – May 2019

- Developed SQL and PL/SQL transformations for data cleansing and analytical processing.
- Automated data security workflows using shell scripting and encryption logic.